MILITARY SPECIFICATION

HOSE, SYNTHETIC RUBBER, WIRE REINFORCED, AND END FITTINGS, REUSABLE, FOR FLEXIBLE HOSE ASSEMBLIES (SINGLE OR DOUBLE WIRE BRAID, MEDIUM PRESSURE SIZES -4 THROUGH -48)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers synthetic rubber hoses reinforced by one or two plies of wire braid in sizes -4 through -48 for medium pressure hydraulic application.

1.2 Classification. End fittings for hoses covered by this specification shall be sizes -4 through -48 as applicable and shall be of the following types:

Type AM - SAE 37-degree tube swivel to hose, straight, hose sizes 1/4 inch to 3 inches (see figure 1).
Type BM - O-ring seal union, male half to hose, straight, hose sizes -6 to -32 (see figure 2).
Type BLM - O-ring seal union, male half to hose, 90-degree elbow, hose sizes -6 to -32 (see figure 3).
Type DM - 90-degree elbow, hose to hose, hose sizes -6 to -32 (see figure 4).
Type FM1 - 150 pounds per square inch (lb/in^2) flange, to hose, straight, hose sizes -16 to -48 (see figure 5).
Type FM11 - 150 lb/in^2 flange, to hose, straight, hose sizes -16 to -48 (see figure 6).
Type FLM1 - 150 lb/in^2 flange, to hose, 90-degree elbow, hose sizes -16 to -48 (see figure 7).
Type FLM11 - 150 lb/in^2 flange, to hose, 90-degree elbow, hose sizes -16 to -48 (see figure 8).
Type GM - Split clamp to hose, straight, with female braze tailpiece and clamp assembly, hose size -32 (see figures 9, 23, 24).
Type GMA - Split clamp to hose, straight, with male braze tailpiece and clamp assembly, hose size -32 (see figures 10, 23, 25).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Type GML - Split clamp to hose, 90-degree elbow, with female braze tailpiece and clamp assembly, hose size -32 (see figures 11, 23, 24).
Type GMLA - Split clamp to hose, 90-degree elbow, with male braze tailpiece and clamp assembly, hose size -32 (see figures 12, 23, 25).
Type AIII - 150 lb/in^2 flange, to hose, straight, hose sizes -40 to -48 (see figure 13).
Type AIV - 150 lb/in^2 flange, to hose, straight, hose sizes -40 to -48 (see figure 14).
Type BIII - 150 lb/in^2 flange, to hose, 90-degree elbow, hose sizes -40 to -48 (see figure 15).
Type BIV - 150 lb/in^2 flange, to hose, 90-degree elbow, hose sizes -40 to -48 (see figure 16).
Type C - Split clamp to hose, straight, with female braze tailpiece and clamp assembly, hose sizes -40 to -48 (see figures 17, 23, 24).
Type CA - Split clamp to hose, straight, with male braze tailpiece and clamp assembly, hose sizes -40 to -48 (see figures 18, 23, 25).
Type D - Split clamp to hose, 90-degree elbow, with female braze tailpiece and clamp assembly, hose sizes -40 to -48 (see figures 19, 23, 24).
Type DA - Split clamp to hose, 90-degree elbow, with male braze tailpiece and clamp assembly, hose sizes -40 to -48 (see figures 20, 23, 25).
Type E - 90-degree elbow fitting, hose to hose, hose sizes -40 to -48 (see figure 21).
Type F - Split clamp fitting, 180-degree elbow, with split clamp assembly, hose to hose, hose sizes -40 to -48 (see figures 22 and 23).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. The following specifications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-C-390 - Copper Alloy Castings (Including Cast Bar).
QQ-C-591 - Copper-Silicon, Copper-Zinc-Silicon, and Copper-Nickel-Silicon Alloys: Rod, Wire, Shapes, Forgings, and Flat Products (Flat Wire, Strip, Sheet, Bar, and Plate).
QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500).
QQ-S-763 - Steel Bars, Wire, Shapes, and Forgings, Corrosion-Resisting.

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MILITARY

MIL-T-1368 - Tube and Pipe, Nickel-Copper Alloy, Seamless and Welded.
MIL-L-2104 - Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-T-16420 - Tube, Copper-Nickel Alloy, Seamless and Welded (Copper Alloy Numbers 715 and 706).
MIL-L-17331 - Lubricating Oil, Steam Turbine and Gear, Moderate Service.
MIL-F-20042 - Flanges, Pipe and Bulkhead, Bronze (Silver Brazing).
MIL-H-24135 - Hose, Synthetic Rubber, Wire Reinforced, and End Fittings, Reusable, for Flexible Hose Assemblies, General Specification for.
MIL-B-24480 - Bronze, Nickel-Aluminum (UNS No. C95800) Castings for Seawater Service.

(Copies of specifications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B16.5 - Pipe Flanges and Flanged Fittings. (DoD adopted)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Hose. Hose furnished under this specification shall conform to the requirements of MIL-H-24135 and as specified herein.

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3.2 Material.

3.2.1 Hose. The rubber compounds used shall be resistant to petroleum base oils and sea water and shall consist of synthetic rubber or a blend of synthetic rubbers and shall meet the requirements of MIL-H-24135 and as specified herein.

3.2.2 Fittings. Material for fittings shall be as shown on figures 1 through 25.

3.3 Reinforcement. The reinforcement shall consist of one or two plies of wire braid applied in such a manner as to meet all the requirements of this specification.

3.4 Dimensions and pressures. Hose dimensions and pressures shall be in accordance with table I.

TABLE I. Dash numbers, performance characteristics, and dimensions.

Hose size (dash no.)	Pressure (lb/in^2)		Hose dimensions (inches)				Min bend radius (inches)	Approx wt/ft (lbs/ft)	Vacuum (in Hg)
	Working (max)	Burst (min)	Id tube		Od cover				
			min	max	min	max			
-4	3,000	12,000	0.188	0.214	0.500	0.539	3.00	0.15	28
-5	2,500	10,000	.250	.281	.562	.601	3.38	.19	28
-6	2,250	9,000	.312	.344	.656	.695	4.00	.23	28
-8	2,000	8,000	.406	.437	.743	.789	4.62	.28	28
-10	1,750	7,000	.500	.539	.899	.945	5.50	.39	28
-12	1,500	6,000	.625	.667	1.055	1.101	6.50	.47	28
-16	800	3,200	.875	.917	1.203	1.265	7.38	.41	20
-20	625	2,500	1.125	1.172	1.469	1.531	9.00	.59	20
-24	500	2,000	1.375	1.422	1.719	1.781	10.50	.72	15
-32	350	1,400	1.812	1.857	2.172	2.266	13.25	.90	11
-40	350	1,400	2.375	2.437	2.828	2.922	24.00	1.63	10
1/ -40	600	2,400	2.484	2.547	3.047	3.141	24.00	2.00	10
-48	200	800	3.00	3.062	3.547	3.641	33.00	2.10	8
1/ -48	450	1,800	3.00	3.062	3.547	3.641	33.00	2.10	8

1/ Applicable when using fittings to figures 13 through 22.

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3.5 Fitting types and sizes. Fitting types and sizes shall be as shown on figures 1 through 22.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with MIL-H-24135 and as specified herein.

4.2 Qualification samples for testing.

4.2.1 Hose. Qualification testing of hose shall consist of hose samples as specified in table II. Samples shall be not less than 15 inches in length and shall be fitted with qualified end fittings.

TABLE II. Numbers of hose samples for qualification testing.

Hose size (dash no.)	Number of hose samples to be provided
-8	4
-32 or 48	5

4.2.2 End fittings. Qualification testing of end fittings only shall consist of 18 end fittings of the same type as specified in table III. End fitting samples shall be installed on nine qualified hose samples, in the sizes listed in table II.

TABLE III. Numbers of end fitting samples for qualification testing.

End fitting size (hose dash no.)	Number of end fitting samples to be provided
-8	8
-32 or 48	10

4.2.3 Hose and end fittings combined. When a single contractor of both hose and fittings qualifies both hose and fittings, or a hose contractor and a fitting contractor qualifies the hose and fittings in a combined qualification test, the samples shall be provided as specified in table IV.

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TABLE IV. Number of samples for combined qualification testing.

One approved source					
Hose size (dash no.)	No. of hose samples to be provided		End fitting (hose dash no.)	No. of end fittings to be provided	
	Qualifier 2/	Approved 1/ source		Qualifier 2/	Approved 1/ source
-8 3/ -32 or -48	2 3	2 2	-8 3/ -32 or -48	4 5	4 5

Two or more approved sources						
Hose size (dash no.)	Number of samples to be provided by			End fitting size (dash no.)	Number of end fittings to be provided by		
	Qualifier 2/	Approved 1/ source no. 1	Approved 1/ source no. 2		Qualifier 2/	Approved 1/ source no. 1	Approved 1/ source no. 2
-8 3/ -32 or -48	2 3	1 1	1 1	-8 3/ -32 or -48	4 5	2 3	2 2

Hose samples shall be not less than 15 inches in length and shall be coupled on one end with the fitting being submitted for qualification and on the other end with previously qualified end fittings.

1/ Approved source - Previously qualified hose and fittings, or hose or fitting.
2/ Qualifier - Hose and fittings, or hose or fitting to be qualified.
3/ When -48 size is offered, -48 size shall be tested.

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4.2.4 Sequence of tests on sample hose assemblies shall be as specified in table V.

TABLE V. Sequence of qualification tests on sample assemblies.

	Sample numbers									
	Size (dash no.) -8				Size (dash no.) -32					-48
Test sequence	1	2	3	4	5	6	7	8	9	10
Proof pressure (see 4.6.1)1/	X	X	X	X	X	X	X	X	X	X
Stability test (see 4.6.2)1/	X	X	X	X	X	X	X	X	X	X
Aging test (see 4.6.3)1/		X	X	X		X	X	X	X	
Hydraulic fluid circulation (see 4.6.4)1/		X				X				
Impulse test (see 4.6.5)1/				X				X	X	
Burst test (see 4.6.6)1/	X				X					
Vacuum test (see 4.6.7)1/										X
Cold temperature flexing (see 4.6.8)1/			X				X			
Adhesion test (see 4.6.9)1/										X

1/ See MIL-H-24135.

4.3 The test fluid shall be petroleum base oil in accordance with MIL-L-17331 or MIL-L-2104 for the aging, hydraulic fluid circulation, and impulse tests.

4.4 The peak pressure for the dynamic impulse test shall be 150 percent of maximum working pressure as specified in table I.

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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 The packaging requirements shall be in accordance with MIL-H-24135.

6. NOTES

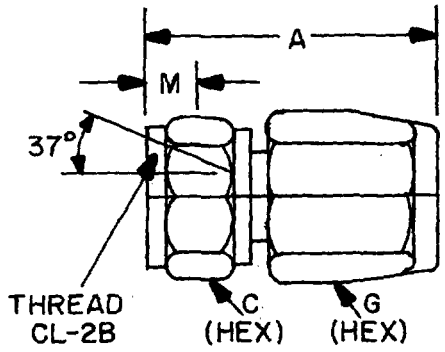
6.1 Intended use. Hose and fittings to this specification are intended for use in shipboard fluid piping systems where pressures do not exceed the rated working pressure of the hose and temperature does not exceed minus 40 to 200°F.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of end fittings for hoses (see 1.2).

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Hose size (dash no.)	Tube size	Thread	A (max)	C (see note 2)	G (see note 2)	M (nom)
-4	1/4	7/16-20	1.75	0.56	0.62	0.35
-5	5/16	1/2-20	1.90	.62	.69	.38
-6	3/8	9/16-18	2.05	.69	.81	.38
-8	1/2	3/4-16	2.52	.88	.94	.43
-10	5/8	7/8-14	2.80	1.00	1.12	.51
-12	3/4	1-1/16-12	3.15	1.25	1.25	.57
-16	1	1-5/16-12	2.82	1.50	1.44	.60
-20	1-1/4	1-5/8-12	3.00	2.00	1.75	.64
-24	1-1/2	1-7/8-12	3.28	2.25	1.94	.74
-32	2-1/2	1-7/8-12	3.88	2.88	2.44	.76
-40	2-1/2	3-12	4.16	4.38	3.25	.80
-48	3	3-1/2-12	4.28	5.00	3.85	.87

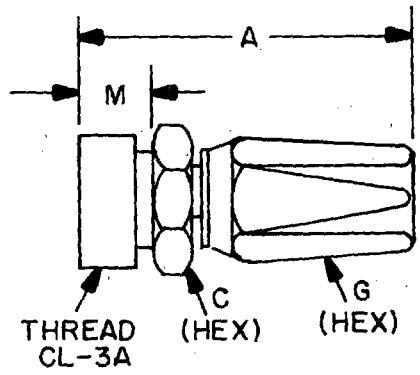
NOTES:

1. Dimensions are in inches.
2. Nut, nipple, and socket shall conform to or be a combination of the following materials (tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286):

   QQ-N-286, form 1 or form 2, condition cold drawn or hot finished.
   QQ-N-281, class A or B, form 1, condition cold drawn.
   QQ-S-763, class 304, condition A or B.
   QQ-C-390, alloy number C90300.

FIGURE 1. Type AM, 37-degree tube swivel to hose, straight, hose sizes 1/4 inch to 3 inches.

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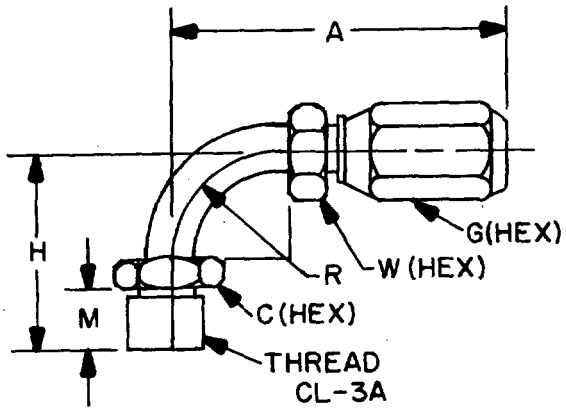
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Hose size (dash no.)	Nps	Thread	A (max)	C (see note 2)	G (see note 2)	M (ref)
-6	1/8	1-14	2.06	1.06	0.81	0.50
-8	1/4	1-3/16-12	2.54	1.25	.94	.62
-10	3/8	1-3/8-12	2.75	1.44	1.12	.62
-12	1/2	1-3/4-12	3.12	1.81	1.35	.62
-16	3/4	2-12	2.80	2.12	1.44	.62
-20	1	2-5/16-12	3.14	2.38	1.75	.75
-24	1-1/4	2-3/4-12	3.23	2.88	1.94	.75
-32	1-1/2	3-1/16-12	3.89	3.12	2.44	.88

NOTES:
1. Dimensions are in inches.
2. Hose nipple material shall conform to QQ-N-286, form 1 or form 2, condition cold drawn or hot finished. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.
3. Socket material shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1 or form 2, condition cold drawn or hot finished.

FIGURE 2. Type BM, O-ring seal union, male half to hose, straight, hose sizes -6 to -32.

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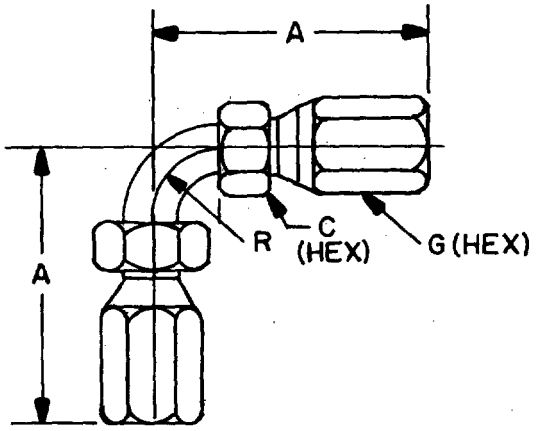
Hose size (dash no.)	Nps	Thread	A (min)	C (see note 2)	G (see note 2)	H (min)	M (ref)	R (max)	W (see note 2)
-6	1/8	1-14	2.45	1.06	0.81	1.30	0.50	0.50	0.56
-8	1/4	1-3/16-12	3.05	1.25	.94	1.68	.62	.75	.75
-10	3/8	1-3/8-12	3.57	1.44	1.12	1.99	.62	1.00	.87
-12	1/2	1-3/4-12	4.15	1.81	1.25	2.3	.62	1.25	1.00
-16	3/4	2-12	3.93	2.12	1.44	2.68	.62	1.50	1.38
-20	1	2-5/16-12	4.27	2.38	1.75	3.17	.75	1.75	1.62
-24	1-1/4	2-3/4-12	4.65	2.88	1.94	3.42	.75	2.00	1.87
-32	1-1/2	3-1/16-12	5.68	3.12	2.44	4.05	.88	2.50	2.38

NOTES:

1. Dimensions are in inches.
2. Hose nipple material shall conform to QQ-N-286, form 1 or form 2, condition cold drawn or hot finished. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.
3. The 90-degree tube elbow for fittings for hose sizes -6 through -12 shall conform to MIL-T-1368, class A; for hose sizes -16 through -32 to MIL-T-16420, type I, grade 2, class 700 special or 1650 special.

FIGURE 3. Type BLM, O-ring seal union, male half to hose, 90-degree elbow, hose sizes -6 to -32.

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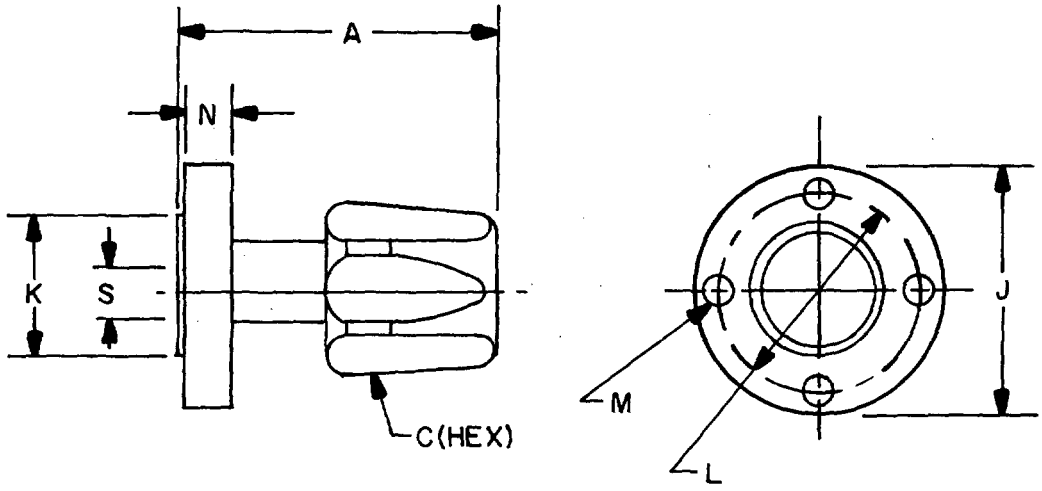
Hose size	A (max)	C (see note 2)	G (see note 2)	R (max)
-6	2.45	0.56	0.81	0.50
-8	3.05	.75	.94	.75
-10	3.57	.81	1.12	1.00
-12	4.15	1.00	1.25	1.25
-16	3.93	1.38	1.44	1.50
-20	2.98	Note	1.75	1.22
-24	3.24	5	1.94	1.38
-32	3.89		2.44	1.62
-40	4.39	-	3.25	1.88

NOTES:

1. Dimensions are in inches.
2. Socket material shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.
3. Hose nipple material shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished; or to QQ-C-390, alloy number C90300 or MIL-B-24480. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.
4. The 90-degree elbow material shall conform to MIL-T-1368, class A; MIL-T-16420, type I, grade 2; class 700 special or 1650 special; QQ-C-390, alloy number C90300; or MIL-B-24480.
5. Sizes -20, -24 and -32 do not have this hex.

FIGURE 4. Type DM, 90-degree elbow, hose to hose, hose sizes -6 to -32.

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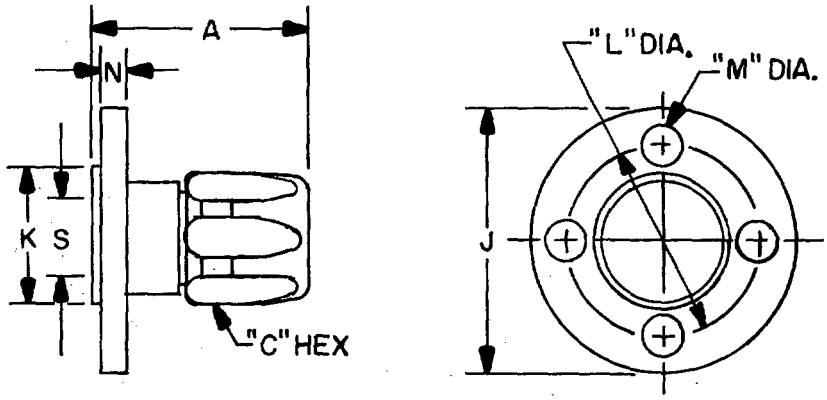
Hose size (dash no.)	Nps	Number of bolt holes	A (max)	C (see note 4)	J (max)	K (max)	L (bolt circle)	M (bolt hole)	N (min)	S (max)
-16	1/2	3	2.80	1.44	3.56	1.50	2.44	0.56	0.38	0.71
-20	3/4	4	2.89	1.75	3.81	1.84	2.69	.56	.44	.92
-24	1	4	2.92	1.94	4.25	2.12	3.12	.56	.44	1.18
-32	1-1/4	4	3.51	2.44	4.50	2.74	3.38	.56	.44	1.53
-32	1-1/2	6	3.51	2.44	5.06	2.74	3.94	.56	.44	0.77
-32	2	6	3.51	2.44	5.56	2.74	4.44	.56	.44	.77
-40	2	6	3.89	3.25	5.56	3.28	4.44	.56	.44	2.24
-40	2-1/2	6	3.89	3.25	6.12	3.28	5.00	.56	.50	2.24
-48	2-1/2	6	3.89	3.88	6.12	3.78	5.00	.56	.50	2.71
-48	3	8	3.89	3.88	6.62	4.50	5.50	.56	.50	3.31

NOTES:

1. Dimensions are in inches.
2. Flange shall be designed to swivel through 360 degrees and dimensioned to mate with 150 lb/in^2 plain flange to MIL-F-20042.
3. Flange and hose nipple material shall conform to QQ-C-390, alloy number C90300 or to MIL-B-24480.
4. Socket material shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1, condition cold drawn or hot finished. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.

FIGURE 5. Type FM1, 150 lb/in^2 flange, to hose, straight, hose sizes -16 to -48.

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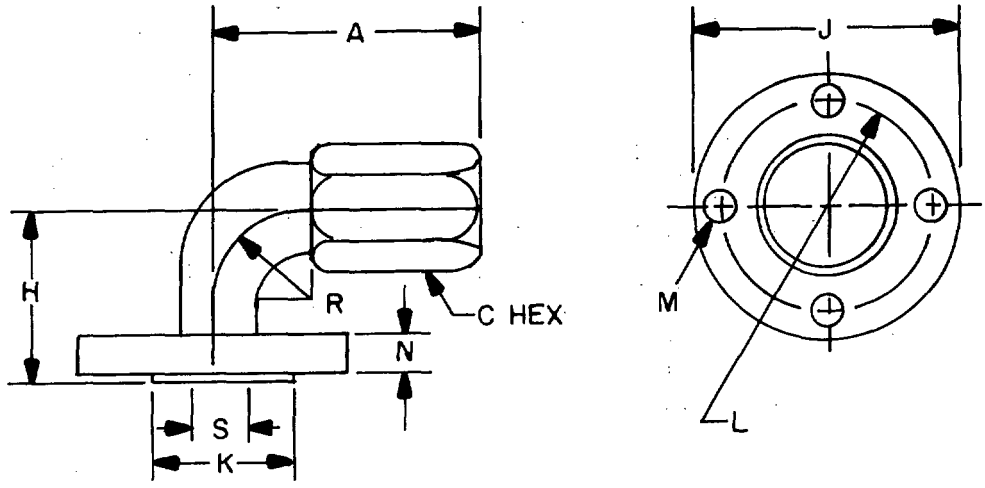
Hose size (dash no.)	Nps	Number of bolt holes	A (max)	C (see note 4)	J (max)	K (max)	L (bolt circle)	M (bolt holes)	N (min)	S (max)
-16	1/2	4	2.80	1.44	3.50	1.50	2.38	0.62	0.58	0.71
-20	3/4	4	2.89	1.75	3.88	1.84	2.75	.62	.44	.92
-24	1	4	2.92	1.94	4.25	2.12	3.12	.62	.44	1.18
-32	1-1/4	4	3.51	2.44	4.62	2.74	3.50	.62	.44	1.53
-32	1-1/2	4	3.51	2.44	5.00	2.74	3.88	.62	.44	1.77
-32	2	4	3.51	2.44	6.00	2.74	4.75	.75	.44	1.77
-40	2	4	3.89	3.25	6.00	3.28	4.75	.75	.44	2.24
-40	2-1/2	4	3.89	3.25	7.00	3.28	5.50	.75	.50	2.24
-48	2-1/2	4	3.89	3.88	7.00	3.78	5.50	.75	.50	2.71

NOTES:

1. Dimensions are in inches.
2. Flange shall be designed to swivel through 360 degrees and dimensioned to mate with ANSI B16.5, class 150 flange.
3. Flange and hose nipple material shall conform to QQ-C-390, alloy number C90300 or to MIL-B-24480.
4. Socket material shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1 or form 2, condition cold drawn or hot finished. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.

FIGURE 6. Type FM11, 150 lb/in^2 flange, to hose, straight, hose sizes -16 to -48.

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SH 1320296

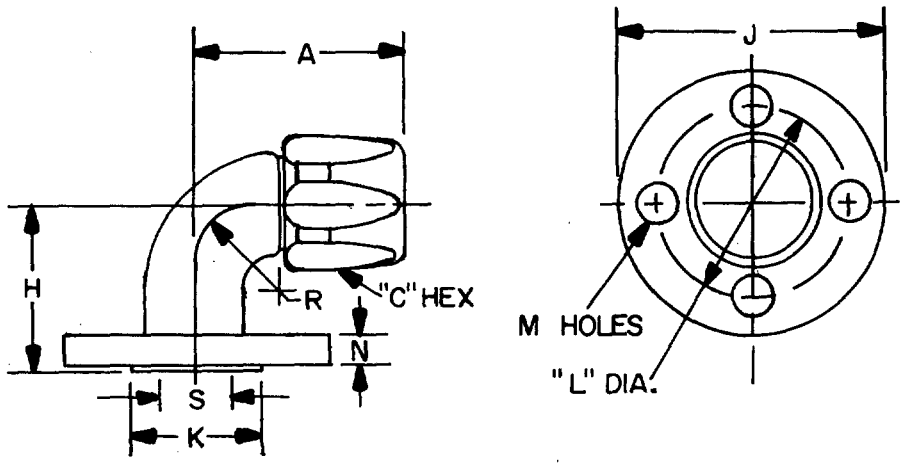
Hose size (dash no.)	Nps	Number of bolt holes	A (max)	C (see note 4)	H (max)	J (max)	K (max)	L (bolt circle)	M (bolt holes)	N (min)	R (nom)	S (max)
-16	1/2	3	2.80	1.44	2.38	3.56	1.50	2.44	0.56	0.38	1.12	0.71
-16	3/4	4	2.80	1.44	2.38	3.81	1.50	2.69	.56	.44	1.12	.71
-20	3/4	4	2.98	1.75	2.56	3.81	1.84	2.69	.56	.44	1.22	.92
-16	1	4	2.80	1.44	2.38	4.25	1.50	3.12	.56	.44	1.12	.71
-24	1	4	3.24	1.94	2.75	4.25	2.12	3.12	.56	.44	1.38	1.18
-32	1-1/4	4	3.89	2.44	3.06	4.50	2.74	3.38	.56	.44	1.62	1.53
-32	1-1/2	6	3.89	2.44	3.06	5.06	2.74	3.94	.56	.44	1.62	1.77
-32	2	6	3.89	2.44	3.06	5.56	2.74	4.44	.56	.44	1.62	1.75

NOTES:

1. Dimensions are in inches.
2. Flange shall be designed to swivel through 360 degrees and be dimensioned to mate with 150 lb/in^2 plain flange to MIL-F-20042.
3. Flange and hose nipple material shall conform to QQ-C-390, alloy number C90300 or to MIL-B-24480.
4. Socket material shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1 or form 2, condition cold drawn or hot finished. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.

FIGURE 7. Type FLM1, 150 lb/in^2 flange, to hose, 90-degree elbow, hose sizes -16 to -48.

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SH 1320297

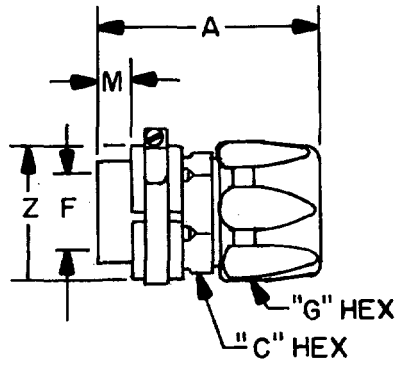
Hose size (dash no.)	Nps	Number of bolt holes	A (max)	C (see note 4)	H (max)	J (max)	K (max)	L (bolt circle)	M (bolt holes)	N (min)	R (nom)	S (max)
-16	1/2	4	2.80	1.44	2.38	3.50	1.50	2.38	0.62	0.38	1.12	0.71
-20	3/4	4	2.98	1.75	2.56	3.88	1.84	2.75	.62	.44	1.22	.92
-24	1	4	3.24	1.94	2.75	4.25	2.12	3.12	.62	.44	1.38	1.18
-32	1-1/4	4	3.89	2.44	3.06	4.62	2.74	3.50	.62	.44	1.62	1.53
-32	1-1/2	4	3.89	2.44	3.06	5.00	2.74	3.88	.62	.44	1.62	1.77
-32	2	4	3.89	2.44	3.06	6.00	2.74	4.75	.75	.44	1.62	1.77

NOTES:

1. Dimensions are in inches.
2. Flange shall be designed to swivel through 360 degrees and be dimensioned to mate with ANSI B16.5, class 150 flange.
3. Flange and hose nipple material shall conform to QQ-C-390, alloy number C90300 or to MIL-B-24480.
4. Socket material shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, form 1 or form 2, condition cold drawn or hot finished. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.

FIGURE 8. Type FLM11, 150 lb/in^2 flange, to hose, 90-degree elbow, hose sizes -16 to -32.

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SH 1320298

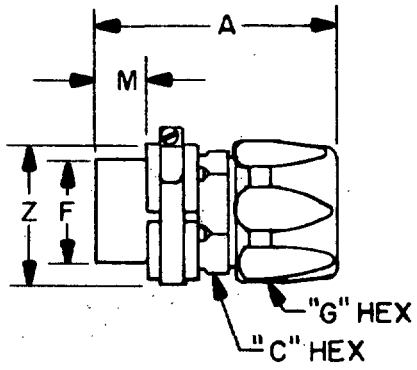
Hose size (dash no.)	Nps	A (max)	C (see note 2)	F (max)	G (see note 2)	M (nom)	Z (nom)
-32	1-1/4	4.98	2.12	1.66	2.44	0.50	3.00
-32	1-1/2	5.03	2.12	1.90	2.44	.62	3.31

NOTES:

1. Dimensions are in inches.
2. Hose nipple material shall conform to QQ-C-390, alloy number C90300 or to MIL-B-24480. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.
3. Socket material shall conform to QQ-N-281, class A or B, condition cold drawn or QQ-N-286, form 1 or form 2, condition cold drawn or hot finished. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.
4. Split clamp assembly shall be to figure 23 as applicable.
5. Silver braze tailpiece shall be to figure 24 as applicable.

FIGURE 9. Type GM, split clamp to hose, straight, hose size -32 (female silver braze adapter).

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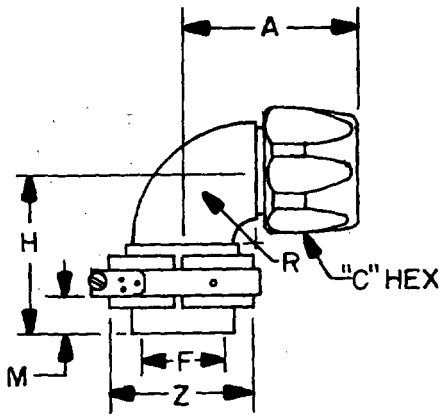
SH 1320299

Hose size (dash no.)	Nps	A (max)	C (see note 2)	F (max)	G (see note 3)	M (nom)	Z (nom)
-32	1-1/4	5.44	2.12	1.66	2.44	1.12	3.00
-32	1-1/2	5.53	2.12	1.90	2.44	1.25	3.31

NOTES:
1. Dimensions are in inches.
2. Hose nipple material shall conform to QQ-C-390, alloy number C90300 or to MIL-B-24480. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.
3. Socket material shall conform to QQ-N-281, class A or B, condition cold drawn or QQ-N-286, form 1 or form 2, condition cold drawn or hot finished. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.
4. Split clamp assembly shall be to figure 23 as applicable.
5. Silver braze tailpiece shall be to figure 25 as applicable.

FIGURE 10. Type GMA, split clamp to hose, straight, hose size -32 (male silver braze adapter).

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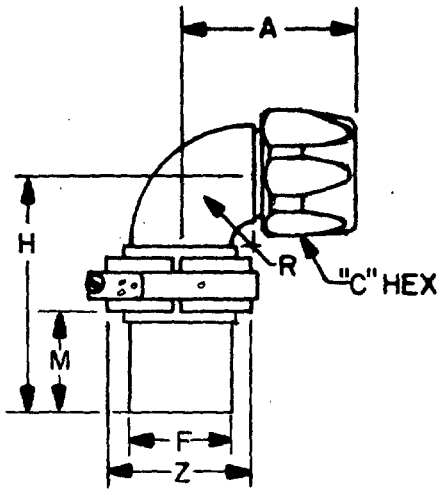
SH 1320300

Hose size (dash no.)	Nps	A (max)	C (see note 3)	F (max)	H (max)	M (nom)	R (nom)	Z (nom)
-32	1-1/4	3.89	2.05	1.66	3.71	0.50	1.62	3.00
-32	1-1/2	3.89	2.05	1.90	3.80	.62	1.62	3.31

NOTES:
1. Dimensions are in inches.
2. Hose nipple material shall conform to QQ-C-390, alloy number C90300 or to MIL-B-24480.
3. Socket material shall conform to QQ-N-281, class A or B, condition cold drawn or QQ-N-286, form 1 or form 2, condition cold drawn or hot finished. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.
4. Split clamp assembly shall be to figure 23 as applicable.
5. Silver braze tailpiece shall be to figure 24 as applicable.

FIGURE 11. Type GML, split clamp to hose, 90-degree elbow, hose size -32 (female silver braze adapter).

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SH 1320301

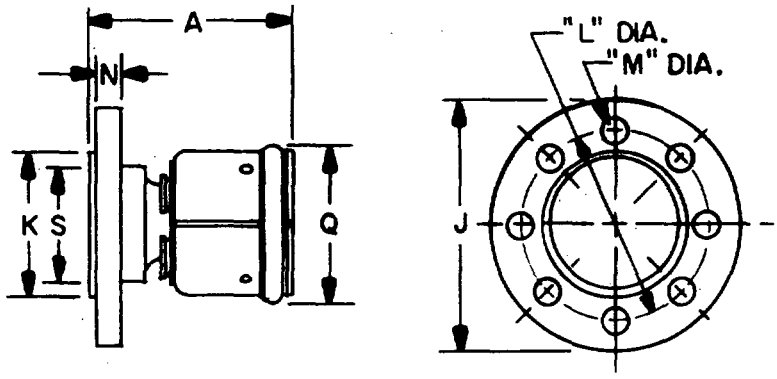
Hose size (dash no.)	Nps	A (max)	C (see note 3)	F (max)	H (max)	M (nom)	R (nom)	Z (nom)
-32	1-1/4	3.89	2.44	1.66	4.18	1.12	1.62	3.00
-32	1-1/2	3.89	2.44	1.90	4.30	1.25	1.62	3.31

NOTES:

1. Dimensions are in inches.
2. Hose nipple material shall conform to QQ-C-390, alloy number C90300 or to MIL-B-24480.
3. Socket material shall conform to QQ-N-281, class A or B, condition cold drawn or QQ-N-286, form 1 or form 2, condition cold drawn or hot finished. Tolerance for the hex dimension shall be determined from the material specification, QQ-N-281 or QQ-N-286.
4. Split clamp assembly shall be to figure 23 as applicable.
5. Silver braze tailpiece shall be to figure 25 as applicable.

FIGURE 12. Type GMLA, split clamp to hose, 90-degree elbow, hose size -32 (male silver braze adapter).

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SH 1320302

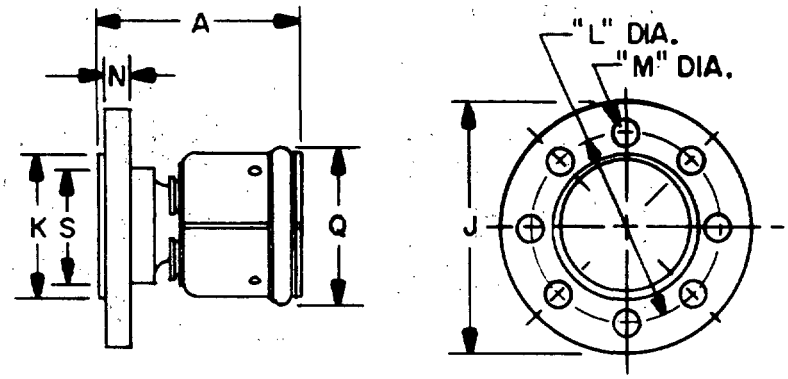
Hose size (dash no.)	Nps	Number of bolt holes	A (max)	J (max)	K (max)	L (bolt circle)	M (bolt hole)	N (min)	Q (nom)	S (max)
-40	2	6	5.16	5.56	3.28	4.44	0.56	0.44	3.90	2.24
-40	2	6	5.16	6.12	3.28	5.00	.56	.50	3.90	2.24
-48	2-1/2	6	5.17	6.12	3.78	5.00	.56	.50	4.50	2.71
-48	3	8	5.17	6.62	4.50	5.50	.56	.50	4.50	3.31

NOTES:

1. Dimensions are in inches.
2. Hose nipple material shall be bronze to MIL-B-24480.
3. Socket assembly may be the multiple segment type, bolt or ring retained, so long as envelope dimensions are not exceeded. Material may be bronze to MIL-B-24480 or conform to QQ-N-281, class A or B or QQ-N-286.
4. Flange material shall be bronze to QQ-C-390, alloy number C90300.
5. Flange shall be designed to swivel through 360 degrees and dimensioned to mate with 400 lb/in^2 plain flange to MIL-F-20042.

FIGURE 13. Type AIII, 150 lb/in^2 flange, to hose, straight, hose sizes -40 to -48.

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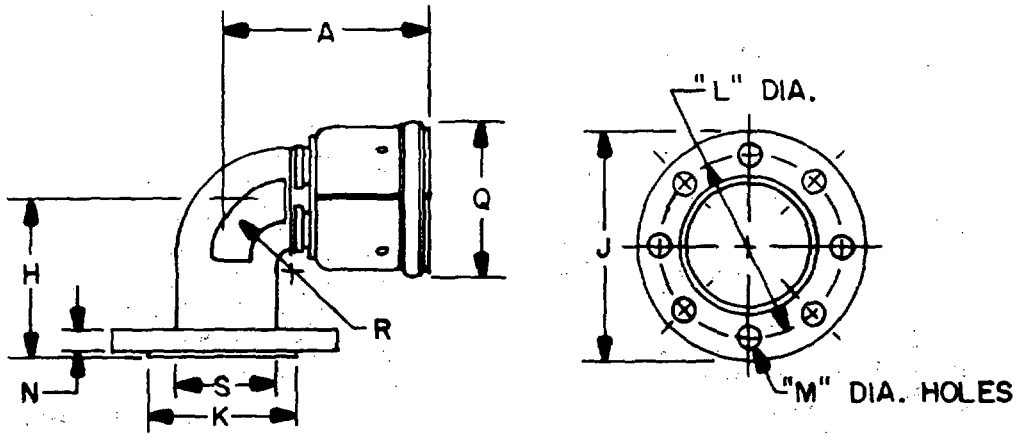
SH 1320303

Hose size (dash no.)	Nps	Number of bolt holes	A (max)	J (max)	K (max)	L (bolt circle)	M (bolt hole)	N (min)	Q (nom)	S (max)
-40	2	4	5.16	6.00	3.28	4.75	0.75	0.44	3.90	2.24
-40	2-1/2	4	5.16	7.00	3.28	5.50	.75	.50	3.90	2.24
-48	2-1/2	4	5.17	7.00	3.78	5.50	.75	.50	4.50	2.71
-48	3	4	5.17	7.50	4.50	6.00	.75	.50	4.50	3.31

NOTES:
1. Dimensions are in inches.
2. Hose nipple material shall be bronze to MIL-B-24480.
3. Socket assembly may be the multiple segment type, bolt or ring retained, so long as envelope dimensions are not exceeded. Material may be bronze to MIL-B-24480 or conform to QQ-N-281, class A or B or QQ-N-286.
4. Flange material shall be bronze to QQ-C-390, alloy number C90300.
5. Flange shall be designed to swivel through 360 degrees and dimensioned to mate with ANSI B16.5, class 150 flange.

FIGURE 14. Type AIV, 150 lb/in^2 flange, to hose, straight, hose sizes -40 to -48.

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SH 1320304

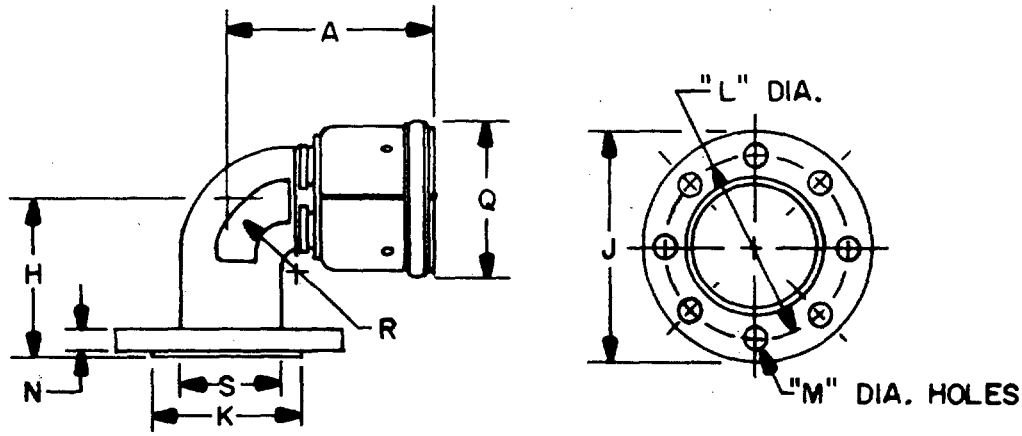
Hose size (dash no.)	Nps	Number of bolt holes	A (max)	H (max)	J (max)	K (max)	L (bolt circle)	M (bolt hole)	N (min)	Q (nom)	R (nom)	S (max)
-40	2	6	5.70	3.50	5.56	3.28	4.44	0.56	0.44	3.90	1.88	2.24
-40	2-1/2	6	5.70	3.50	6.12	3.28	5.00	.56	.50	3.90	1.88	2.24
-48	2-1/2	6	5.96	4.00	6.12	3.78	5.00	.56	.50	4.50	2.09	2.71
-48	3	6	5.96	4.00	6.62	4.50	5.50	.56	.50	4.50	2.09	3.31

NOTES:

1. Dimensions are in inches.
2. Hose nipple material shall be bronze to MIL-B-24480.
3. Socket assembly may be the multiple segment type, bolt or ring retained, so long as envelope dimensions are not exceeded. Material may be bronze to MIL-B-24480 or conform to QQ-N-281, class A or B or QQ-N-286.
4. Flange material shall be bronze to QQ-C-390, alloy number C90300.
5. Flange shall be designed to swivel through 360 degrees and dimensioned to mate with 150 lb/in^2 plain flange to MIL-F-20042.

FIGURE 15. Type BIII, 150 lb/in^2 flange, to hose, 90-degree elbow, hose sizes -40 to -48.

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SH 1320305

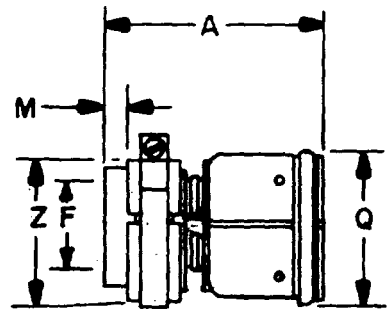
Hose size (dash no.)	Nps	Number of bolt holes	A (max)	H (max)	J (max)	K (max)	L (bolt circle)	M (bolt hole)	N (min)	Q (nom)	R (nom)	S (max)
-40	2	4	5.70	3.50	6.00	3.28	4.75	0.75	0.44	3.90	1.88	2.24
-40	2-1/2	4	5.70	3.50	7.00	3.28	5.50	.75	.50	3.90	1.88	2.24
-48	2-1/2	4	5.96	4.00	7.00	3.78	5.50	.75	.50	4.50	2.09	2.71
-48	3	4	5.96	4.00	7.50	4.50	6.00	.75	.50	4.50	2.09	3.31

NOTES:

1. Dimensions are in inches.
2. Hose nipple material shall be bronze to MIL-B-24480.
3. Socket assembly may be the multiple segment type, bolt or ring retained, so long as envelope dimensions are not exceeded. Material may be bronze to MIL-B-24480 or conform to QQ-N-281, class A or B or QQ-N-286.
4. Flange material shall be bronze to QQ-C-390, alloy number C90300.
5. Flange shall be designed to swivel through 360 degrees and dimensioned to mate with ANSI B16.5, class 150 flange.

FIGURE 16. Type BIV, 150 lb/in^2 flange, to hose, 90-degree elbow, hose sizes -40 to -48.

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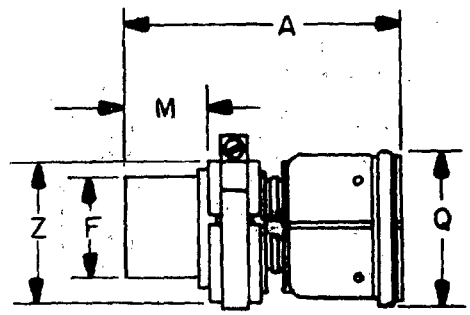
Hose size (dash no.)	Nps	A (max)	F (max)	M (nom)	Q (nom)	Z (nom)
-40	2	6.10	2.38	0.66	3.90	4.00
-48	2-1/2	6.42	2.88	.78	4.50	4.42
-48	3	6.11	3.50	.83	4.50	5.19

NOTES:

1. Dimensions are in inches.
2. Hose nipple material shall be bronze to MIL-B-24480.
3. Socket assembly may be the multiple segment type, bolt or ring retained, so long as envelope dimensions are not exceeded. Material may be bronze to MIL-B-24480 or conform to QQ-N-281, class A or B or QQ-N-286.
4. Split clamp assembly shall be to figure 15 as applicable.
5. Silver braze tailpiece shall be to figure 16 as applicable.

FIGURE 17. Type C, split clamp to hose, straight, hose sizes -40 to -48.

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SH 1320307

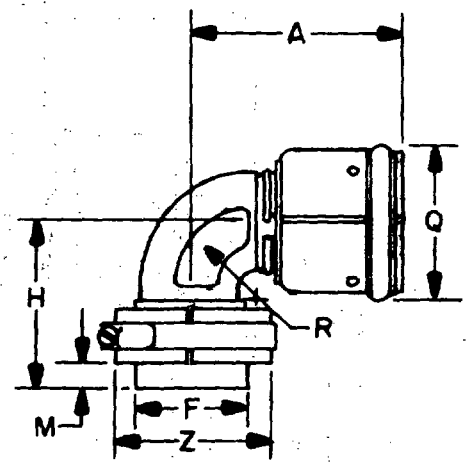
Hose size (dash no.)	Nps	A (max)	F (max)	M (nom)	Q (nom)	Z (nom)
-40	2	6.61	2.38	1.31	3.90	4.00
-48	2-1/2	6.60	2.88	1.12	4.50	4.42
-48	3	8.01	3.50	2.50	4.50	5.19

NOTES:

1. Dimensions are in inches.
2. Hose nipple material shall be bronze to MIL-B-24480.
3. Socket assembly may be the multiple segment type, bolt or ring retained, so long as envelope dimensions are not exceeded. Material shall conform to QQ-N-281, class A or B or QQ-N-286, or be bronze to MIL-B-24480.
4. Split clamp assembly shall be to figure 15 as applicable.
5. Silver braze tailpiece shall be to figure 17 as applicable.

FIGURE 18. Type CA, split clamp to hose, straight, hose sizes -40 to -48.

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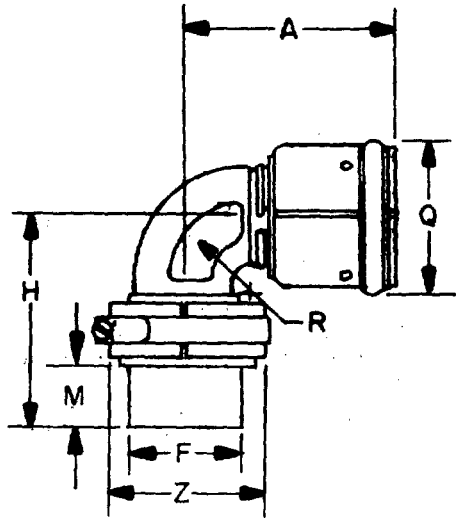
SH 1320308

Hose size (dash no.)	Nps	A (max)	F (max)	H (max)	M (nom)	Q (nom)	R (nom)	Z (nom)
-40	2	5.70	2.38	3.88	0.66	3.90	1.99	4.00
-48	2-1/2	5.88	2.88	4.51	.78	4.50	2.09	4.42
-48	3	6.46	3.50	4.25	.83	4.50	2.09	5.19

NOTES:
1. Dimensions are in inches.
2. Hose nipple material shall be bronze to MIL-B-24480.
3. Socket assembly may be the multiple segment type, bolt or ring retained, so long as envelope dimensions are not exceeded. Material may be bronze to MIL-B-24480 or conform to QQ-N-281, class A or B or QQ-N-286.
4. Split clamp assembly shall be to figure 15 as applicable.
5. Silver braze tailpiece shall be to figure 16 as applicable.

FIGURE 19. Type D, split clamp to hose, 90-degree elbow, hose sizes -40 to -48.

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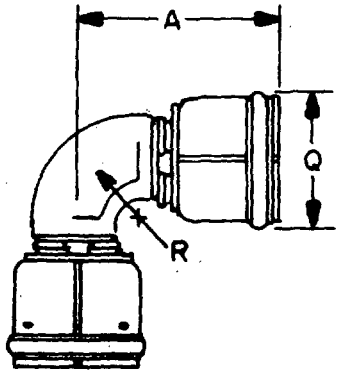
SH 1320309

Hose size (dash no.)	Nps	A (max)	F (max)	H (max)	M (nom)	Q (nom)	R (nom)	Z (nom)
-40	2	5.70	2.38	3.88	0.66	3.90	1.99	4.00
-48	2-1/2	5.88	2.88	4.51	.78	4.50	2.09	4.42
-48	3	6.46	3.50	4.25	.83	4.50	2.09	5.19

NOTES:
1. Dimensions are in inches.
2. Hose nipple material shall be bronze to MIL-B-24480.
3. Socket assembly may be the multiple segment type, bolt or ring retained, so long as envelope dimensions are not exceeded. Material may be bronze to MIL-B-24480 or conform to QQ-N-281, class A or B or QQ-N-286.
4. Split clamp assembly shall be to figure 15 as applicable.
5. Silver braze tailpiece shall be to figure 17 as applicable.

FIGURE 20. Type DA, split clamp to hose, 90-degree elbow, hose sizes -40 to -48.

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SH 1320310

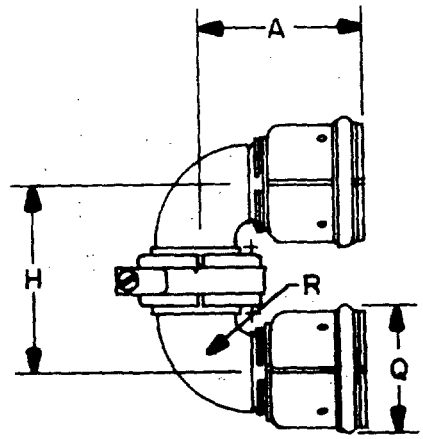
Hose size (dash no.)	A (max)	Q (nom)	R (nom)
-40	5.70	3.90	1.88
-48	5.96	4.50	2.09

NOTES:

1. Dimensions are in inches.
2. Hose nipple material shall be bronze to MIL-B-24480.
3. Socket assembly may be the multiple segment type, bolt or ring retained, so long as envelope dimensions are not exceeded. Material may be bronze to MIL-B-24480 or conform to QQ-N-281, class A or B or QQ-N-286.

FIGURE 21. Type E, 90-degree elbow fitting, hose to hose, hose sizes -40 to -48.

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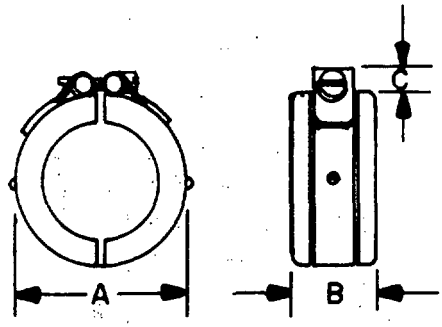
SH 1320311

Hose size (dash no.)	Nps	A (max)	H (max)	Q (nom)	R (nom)
-40	2	5.70	5.40	3.90	1.88
-48	2-1/2	6.05	5.92	4.50	2.09
-48	3	6.46	5.95	4.50	2.09

NOTES:
1. Dimensions are in inches.
2. Hose nipple material shall be bronze to MIL-B-24480.
3. Socket assembly may be the multiple segment type, bolt or ring retained, so long as envelope dimensions are not exceeded. Material may be bronze to MIL-B-24480 or conform to QQ-N-281, class A or B or QQ-N-286.
4. Split clamp assembly shall be to figure 15 as applicable.

FIGURE 22. Type F, split clamp fitting, 180-degree elbow, hose sizes -40 to -48.

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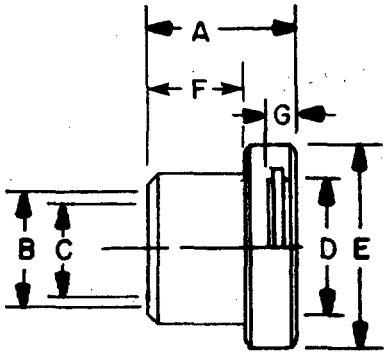
SH 1320312

Nps	A (nom)	B (max)	C (nom)
2	4.00	1.38	0.39
2-1/2	4.42	1.44	.39
3	5.19	1.50	.39

NOTES:
1. Dimensions are in inches.
2. Split clamp halves shall be bronze to QQ-C-390, alloy number C95500.
3. Screws and nuts shall be bronze conforming to QQ-C-591, alloy number 651 or 655, or QQ-N-281.
4. Band shall be nickel-copper to QQ-N-281.

FIGURE 23. Split clamp assembly.

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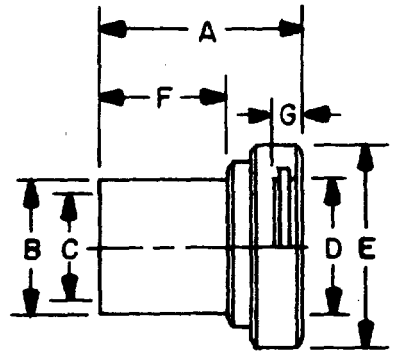
SH 1320313

Nps	A (max)	B (max)	C (max)	D (max)	E (max)	F (max)	G (max)
2	1.64	2.38	2.12	2.51	3.25	0.66	0.35
2-1/2	1.86	2.88	2.59	3.01	3.75	.78	.35
3	1.94	3.50	3.17	3.64	4.36	.83	.35

NOTES:
1. Dimensions are in inches.
2. Material for silver braze tailpiece shall be bronze to QQ-C-390, alloy number C90300 or MIL-B-24480.
3. Braze socket and groove shall be designed for ultrasonic inspection.
4. Back-up ring shall be commercial grade leather.
5. O-ring shall be synthetic rubber suitable for system service.

FIGURE 24. Female silver braze tailpiece.

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SH 1320314

Nps	A (max)	B (max)	C (max)	D (max)	E (max)	F (max)	G (max)
2	2.17	2.38	2.04	2.51	3.24	1.31	0.35
2-1/2	2.04	2.88	2.59	3.01	3.74	1.12	.35
3	3.45	3.50	3.00	3.63	4.37	2.50	.35

NOTES:
1. Dimensions are in inches.
2. Material for silver braze tailpiece shall be bronze to QQ-C-390, alloy number C90300 or MIL-B-24480.
3. Back-up ring shall be commercial grade leather.
4. O-ring shall be synthetic rubber suitable for system service.

FIGURE 25. Male silver braze tailpiece.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-24135/10(SH)	Hose, Synthetic Rubber, Wire Reinforced, And End Fittings, Reusable, For Flexible Hose Assemblies

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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